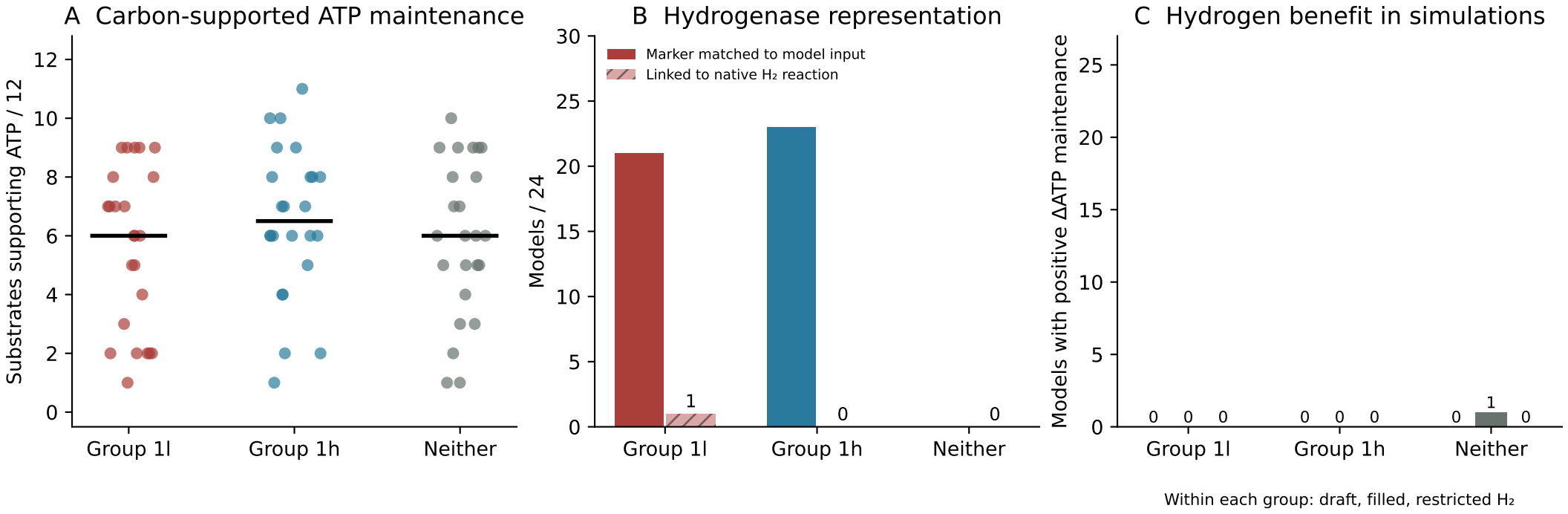
A Carbon-supported ATP maintenance
Substrates supporting ATP / 12
0
2
4
6
8
10
12
Group 1l
Group 1h
Neither
B Hydrogenase representation
Models / 24
0
5
10
15
20
25
30
Marker matched to model input
Linked to native H₂ reaction
1
0
0
Group 1l
Group 1h
Neither
C Hydrogen benefit in simulations
Models with positive ΔATP maintenance
0
5
10
15
20
25
0 0 0
0 0 0
0 1 0
Group 1l
Group 1h
Neither
Within each group: draft, filled, restricted H₂